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Killer Rigs

Want to build a system but don't know what components to buy? Read on to know more.



Pick one that suits the use

Selecting the right processor is not as hard as it sounds, you just need to get a grip on what you will use it for and then proceed



Agent 001

agent001@thinkdigit.com

While going through all the letters we receive, requests for computer/laptop configurations are one of the most popular queries we receive. People need a processor that fits within a particular budget (mostly meagre) and should be able to process everything under the sun. Unfortunately, things of that calibre are yet to be viable for the consumer market. Who am I kidding? Those things will never hit the consumer market. Before it does so, it'll be stripped down and different SKUs of the same die will be brought out. Each with a unique feature suiting a particular budget range and given the popular trend, each with its own unique socket. Manufacturers have their own agenda for minting the most money it can while holding off innovation.

Choosing a processor is no biggie, all you got to do is list out what all you need it to do. Once you've got that taken care of, visit benchmarking websites and figure out which processor can handle which task the best. After you've shortlisted all the processors, you need to figure out which processor can handle most of the tasks you've listed. Chances are that within your budget you'll find a lot of processors. However, being in India, you will rarely have access to all of them.

If you want a budget system, then you'll be recommended the Pentium G620 or an AMD A6-3650 for integrated graphics. It's not because these processors are the best

for you but because that's all vendors stock. Did you know that for an absolutely budget friendly PC, a G530 does the job that a G620 can while knocking off a few hundred rupees? Or that a G850 which is a slightly expensive processor can be underclocked to reduce the overall power consumption while still being capable of doing more than the G620? Whenever, you feel more processing power is needed, bring it back to the stock frequency and you have a better processor. It works out to be a much viable option for long term budget builds.

Processors which can handle gaming needs a greater frequency range as some games are not that dependent upon the processor while some such as *Grand Theft Auto 4* and *Civilization 5* are. At low resolutions, in most games, the CPU becomes the bottleneck. However, most games aren't capable of making the best of a multi-threaded processor. So the usual recommendations end up being the Sandy Bridge i5-2300/2400/2500/2500K. The 2500K being the most versatile and most expensive among them. It's been months since Ivy Bridge has entered the market but when considering the cost, the Sandy Bridge processors are the way to go. Especially since the Sandy Bridge processors overclock much better and hold at lower temperatures. The iGPU on Ivy Bridge is better but when you consider the processor for a gaming machine, it becomes redundant since these machines will have a dedicated graphics card which will outperform the iGPU by leaps and bounds. When Intel comes out with a new processor after 18

months all you need to do is raise the multiplier up by a few notches and you'll have a processor that works at par.

For the top segment, the candidates hold enough power to make a machine that will last 6-8 years when raw processing power is needed. The premium segment doesn't show much advantage over the i5 for gaming. However, pure multi-threaded applications such as modelling/design software (Blender/3Ds Max/Maya), server applications and programming IDEs benefit a lot from a multi-threaded processor. So much that these applications have long since started focussing on the parallel processing architecture found on GPUs rather than CPUs. It won't be long before they are completely GPU dependent. For example, this month we performed GPU benchmarks and found that a simple AES encryption or decryption iteration performs better on the CPU, but when 1000 iterations of the same are to be performed, the CPU doesn't even stand a chance. Server processors are best suited for the job, but they're designed for long term use at a constant load and come with instruction sets for server based applications. Using such a processor for gaming is not unheard of but won't match the performance you'd get with the Core i7 series. The premium segment from Intel/AMD provides the immense computational power of each architecture while keeping it within a price point which is favourable that's what makes it appealing over Xeon/Opteron processors. Money is what decides whether you go in for a Core-i7 or a configuration that involves 1-4 Xeons on a workstation board. 